

Land Survey

Engineering Mathematics III for Land Survey

Professional Learner Notes

Survey statistics, probability, matrices, errors and adjustment concepts for higher-level survey mathematics.

Section	Purpose
Survey data collection	Repeated observations and data quality.
Central tendency	Mean, median and mode for observations.
Dispersion	Range, variance and standard deviation.
Probability	Simple probability and quality decision-making.
Matrices and adjustment	Linear systems and survey corrections.

How to Study This Topic

The learning method follows a programmed engineering mathematics approach: understand the concept, study the formula, follow a worked method, then practise with a technical problem. The aim is not memorisation only; the learner must interpret the answer in the department context.

- Write the formula before substitution.
- Use correct units.
- Check whether the result is reasonable.
- State a technical interpretation.
- Repeat practice until the method is fluent.

Formula Summary

Mean

$$\bar{x} = \frac{\sum x}{n}$$

Meaning: identify each variable, check the units, substitute carefully, and state the final answer in the correct engineering context.

Standard deviation

$$s = \sqrt{\frac{\sum (x - \bar{x})^2}{n-1}}$$

Meaning: identify each variable, check the units, substitute carefully, and state the final answer in the correct engineering context.

Probability

$$P(A) = \frac{n(A)}{n(S)}$$

Meaning: identify each variable, check the units, substitute carefully, and state the final answer in the correct engineering context.

Matrix system

$$AX = B$$

Meaning: identify each variable, check the units, substitute carefully, and state the final answer in the correct engineering context.

Error

$$e = \text{observed} - \text{true}$$

Meaning: identify each variable, check the units, substitute carefully, and state the final answer in the correct engineering context.

1. Survey data collection

Repeated observations and data quality.

In land survey, this section is important because trainees use the calculation to make practical decisions in drawings, site measurements, setting out, material estimation, survey observations, quality checks, or revision work.

Key learning points

- Identify the physical quantity being calculated.
- Separate given values from required values.
- Select a formula that matches the problem.
- Use consistent units before calculation.
- Interpret the result as a technician, not only as a number.

Main formula for this section

$$\bar{x} = \frac{\sum x}{n}$$

Worked method

Example: Repeated angles. Calculate mean of repeated angular observations.

Step 1: Write the known quantities. Step 2: Write the formula. Step 3: Substitute the values. Step 4: simplify carefully. Step 5: attach units. Step 6: write a one-sentence technical interpretation.

Engineering application

This method supports repeated observations. The result can be used to check whether a site measurement, drawing interpretation, material quantity, or survey computation is technically reasonable.

Common learner errors

- Using the wrong formula because the diagram/context was not analysed.
- Mixing units such as mm, cm and m.
- Rounding too early.
- Losing signs or angle direction.
- Submitting an answer without interpretation.

Practice set

1. Create and solve one technical calculation involving survey adjustment. Show formula, substitution, units and interpretation.
2. Create and solve one technical calculation involving quality checks. Show formula, substitution, units and interpretation.
3. Create and solve one technical calculation involving probability of acceptance. Show formula, substitution, units and interpretation.
4. Create and solve one technical calculation involving repeated observations. Show formula, substitution, units and interpretation.
5. Create and solve one technical calculation involving measurement errors. Show formula, substitution, units and interpretation.

6. Create and solve one technical calculation involving survey adjustment. Show formula, substitution, units and interpretation.
7. Create and solve one technical calculation involving quality checks. Show formula, substitution, units and interpretation.
8. Create and solve one technical calculation involving probability of acceptance. Show formula, substitution, units and interpretation.

2. Central tendency

Mean, median and mode for observations.

In land survey, this section is important because trainees use the calculation to make practical decisions in drawings, site measurements, setting out, material estimation, survey observations, quality checks, or revision work.

Key learning points

- Identify the physical quantity being calculated.
- Separate given values from required values.
- Select a formula that matches the problem.
- Use consistent units before calculation.
- Interpret the result as a technician, not only as a number.

Main formula for this section

$$s = \sqrt{\frac{\sum (x - \bar{x})^2}{n-1}}$$

Worked method

Example: Observation spread. Calculate standard deviation of distance measurements.

Step 1: Write the known quantities. Step 2: Write the formula. Step 3: Substitute the values.

Step 4: simplify carefully. Step 5: attach units. Step 6: write a one-sentence technical interpretation.

Engineering application

This method supports measurement errors. The result can be used to check whether a site measurement, drawing interpretation, material quantity, or survey computation is technically reasonable.

Common learner errors

- Using the wrong formula because the diagram/context was not analysed.
- Mixing units such as mm, cm and m.
- Rounding too early.
- Losing signs or angle direction.
- Submitting an answer without interpretation.

Practice set

1. Create and solve one technical calculation involving quality checks. Show formula, substitution, units and interpretation.
2. Create and solve one technical calculation involving probability of acceptance. Show formula, substitution, units and interpretation.
3. Create and solve one technical calculation involving repeated observations. Show formula, substitution, units and interpretation.
4. Create and solve one technical calculation involving measurement errors. Show formula, substitution, units and interpretation.
5. Create and solve one technical calculation involving survey adjustment. Show formula, substitution, units and interpretation.

6. Create and solve one technical calculation involving quality checks. Show formula, substitution, units and interpretation.
7. Create and solve one technical calculation involving probability of acceptance. Show formula, substitution, units and interpretation.
8. Create and solve one technical calculation involving repeated observations. Show formula, substitution, units and interpretation.

3. Dispersion

Range, variance and standard deviation.

In land survey, this section is important because trainees use the calculation to make practical decisions in drawings, site measurements, setting out, material estimation, survey observations, quality checks, or revision work.

Key learning points

- Identify the physical quantity being calculated.
- Separate given values from required values.
- Select a formula that matches the problem.
- Use consistent units before calculation.
- Interpret the result as a technician, not only as a number.

Main formula for this section

$$P(A) = \frac{n(A)}{n(S)}$$

Worked method

Example: Matrix adjustment. Represent correction equations in matrix form.

Step 1: Write the known quantities. Step 2: Write the formula. Step 3: Substitute the values. Step 4: simplify carefully. Step 5: attach units. Step 6: write a one-sentence technical interpretation.

Engineering application

This method supports survey adjustment. The result can be used to check whether a site measurement, drawing interpretation, material quantity, or survey computation is technically reasonable.

Common learner errors

- Using the wrong formula because the diagram/context was not analysed.
- Mixing units such as mm, cm and m.
- Rounding too early.
- Losing signs or angle direction.
- Submitting an answer without interpretation.

Practice set

1. Create and solve one technical calculation involving probability of acceptance. Show formula, substitution, units and interpretation.
2. Create and solve one technical calculation involving repeated observations. Show formula, substitution, units and interpretation.
3. Create and solve one technical calculation involving measurement errors. Show formula, substitution, units and interpretation.
4. Create and solve one technical calculation involving survey adjustment. Show formula, substitution, units and interpretation.
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6. Create and solve one technical calculation involving probability of acceptance. Show formula, substitution, units and interpretation.
7. Create and solve one technical calculation involving repeated observations. Show formula, substitution, units and interpretation.
8. Create and solve one technical calculation involving measurement errors. Show formula, substitution, units and interpretation.

4. Probability

Simple probability and quality decision-making.

In land survey, this section is important because trainees use the calculation to make practical decisions in drawings, site measurements, setting out, material estimation, survey observations, quality checks, or revision work.

Key learning points

- Identify the physical quantity being calculated.
- Separate given values from required values.
- Select a formula that matches the problem.
- Use consistent units before calculation.
- Interpret the result as a technician, not only as a number.

Main formula for this section

$$AX=B$$

Worked method

Example: Repeated angles. Calculate mean of repeated angular observations.

Step 1: Write the known quantities. Step 2: Write the formula. Step 3: Substitute the values. Step 4: simplify carefully. Step 5: attach units. Step 6: write a one-sentence technical interpretation.

Engineering application

This method supports quality checks. The result can be used to check whether a site measurement, drawing interpretation, material quantity, or survey computation is technically reasonable.

Common learner errors

- Using the wrong formula because the diagram/context was not analysed.
- Mixing units such as mm, cm and m.
- Rounding too early.
- Losing signs or angle direction.
- Submitting an answer without interpretation.

Practice set

1. Create and solve one technical calculation involving repeated observations. Show formula, substitution, units and interpretation.
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5. Create and solve one technical calculation involving probability of acceptance. Show formula, substitution, units and interpretation.

6. Create and solve one technical calculation involving repeated observations. Show formula, substitution, units and interpretation.
7. Create and solve one technical calculation involving measurement errors. Show formula, substitution, units and interpretation.
8. Create and solve one technical calculation involving survey adjustment. Show formula, substitution, units and interpretation.

5. Matrices and adjustment

Linear systems and survey corrections.

In land survey, this section is important because trainees use the calculation to make practical decisions in drawings, site measurements, setting out, material estimation, survey observations, quality checks, or revision work.

Key learning points

- Identify the physical quantity being calculated.
- Separate given values from required values.
- Select a formula that matches the problem.
- Use consistent units before calculation.
- Interpret the result as a technician, not only as a number.

Main formula for this section

$$e = \mathrm{observed} - \mathrm{true}$$

Worked method

Example: Observation spread. Calculate standard deviation of distance measurements.

Step 1: Write the known quantities. Step 2: Write the formula. Step 3: Substitute the values.

Step 4: simplify carefully. Step 5: attach units. Step 6: write a one-sentence technical interpretation.

Engineering application

This method supports probability of acceptance. The result can be used to check whether a site measurement, drawing interpretation, material quantity, or survey computation is technically reasonable.

Common learner errors

- Using the wrong formula because the diagram/context was not analysed.
- Mixing units such as mm, cm and m.
- Rounding too early.
- Losing signs or angle direction.
- Submitting an answer without interpretation.

Practice set

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8. Create and solve one technical calculation involving quality checks. Show formula, substitution, units and interpretation.

Mixed Revision Questions

1. A land survey problem involves measurement errors. Choose a suitable mathematical method, solve the problem, and explain the result.
2. A land survey problem involves survey adjustment. Choose a suitable mathematical method, solve the problem, and explain the result.
3. A land survey problem involves quality checks. Choose a suitable mathematical method, solve the problem, and explain the result.
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29. A land survey problem involves probability of acceptance. Choose a suitable mathematical method, solve the problem, and explain the result.
30. A land survey problem involves repeated observations. Choose a suitable mathematical method, solve the problem, and explain the result.

References and Learning Sources

These learner materials are original notes prepared for Mr Mwirigi Mathematics Training Hub. The topic sequencing and level are informed by standard engineering mathematics references and the uploaded curriculum/OS materials. No textbook pages or solution-manual pages have been reproduced.

1. John Bird, Engineering Mathematics, Newnes / Elsevier.
2. K. A. Stroud and Dexter J. Booth, Engineering Mathematics, Palgrave Macmillan.
3. Uploaded KNEC solved mathematics materials for examination-style orientation.
4. Civil Engineering, Building Technology and Land Survey curriculum and occupational-standard materials used for local alignment.